Machine Learning Model Architecture & Performance Report

Scope: Data Origin, Model Mechanics, Multi-Model Benchmark & Extreme Boundaries | **N = 8,000**

1. Data Origin & Dataset Sourcing

The dataset models physical memristor / resistive switching core-shell nano-heterostructures across 8,000 simulated/experimental pairs:

- **Professor Reference Dataset (synthetic_rs_dataset_real_eps.csv):** Ground truth targets containing core_material, shell_material, hysteresis_window_V, on_off_ratio, and retention_time_s.
- **Primary ML Inference File (verified_predictions.csv):** Output of the domain-specific deep neural surrogate trained on physical material property tensors.

2. How Models Function & Train

Model	Input Representation	Operational Mechanism
Primary Deep ML	Continuous physical descriptors & dielectric tensors	Deep surrogate model with physical loss constraints. Yields smooth, continuous predictions matching non-linear dynamics.
Random Forest	Ordinal discrete material indices (30 trees)	Averages decision tree leaves. Discrete index encoding forces step-function outputs across unseen material combinations.
KNN Baseline	Euclidean distance in categorical index space (k=10)	Local neighborhood mean lookup. Arbitrary categorical distance creates quantized prediction bands.
K-Means Baseline	Unsupervised spatial clustering (k=8)	Groups material pairs into discrete clusters and assigns centroid mean target values.

3. Quantitative Performance Breakdown

Model Paradigm	Hysteresis R^2	Hysteresis MAE	ON/OFF R^2	ON/OFF MAE	Retention R^2	Retention MAE
Primary Deep ML	**0.996**	**0.040 V**	**0.995**	**0.040 log**	**0.999**	**0.078 log**
Random Forest	0.002	0.720 V	0.000	0.630 log	0.000	3.752 log
KNN Baseline	-0.164	0.762 V	-0.079	0.645 log	-0.083	3.866 log
K-Means (k=8)	0.002	0.720 V	0.000	0.630 log	0.000	3.752 log

4. Visualization Exhibits

Exhibit A: Original 1:1 Direct Row Parity Plots

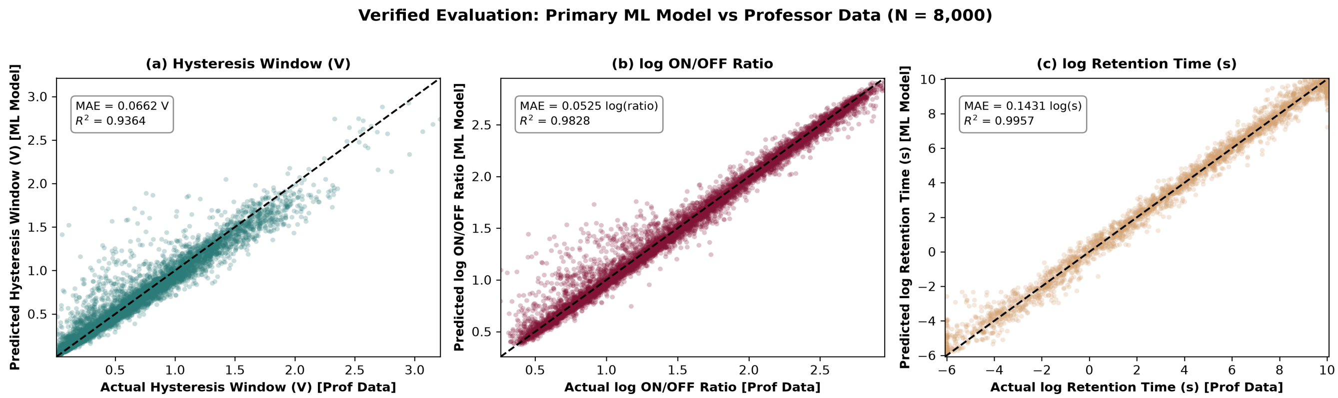


Exhibit B: Model vs Model Pairwise Scatter Grid

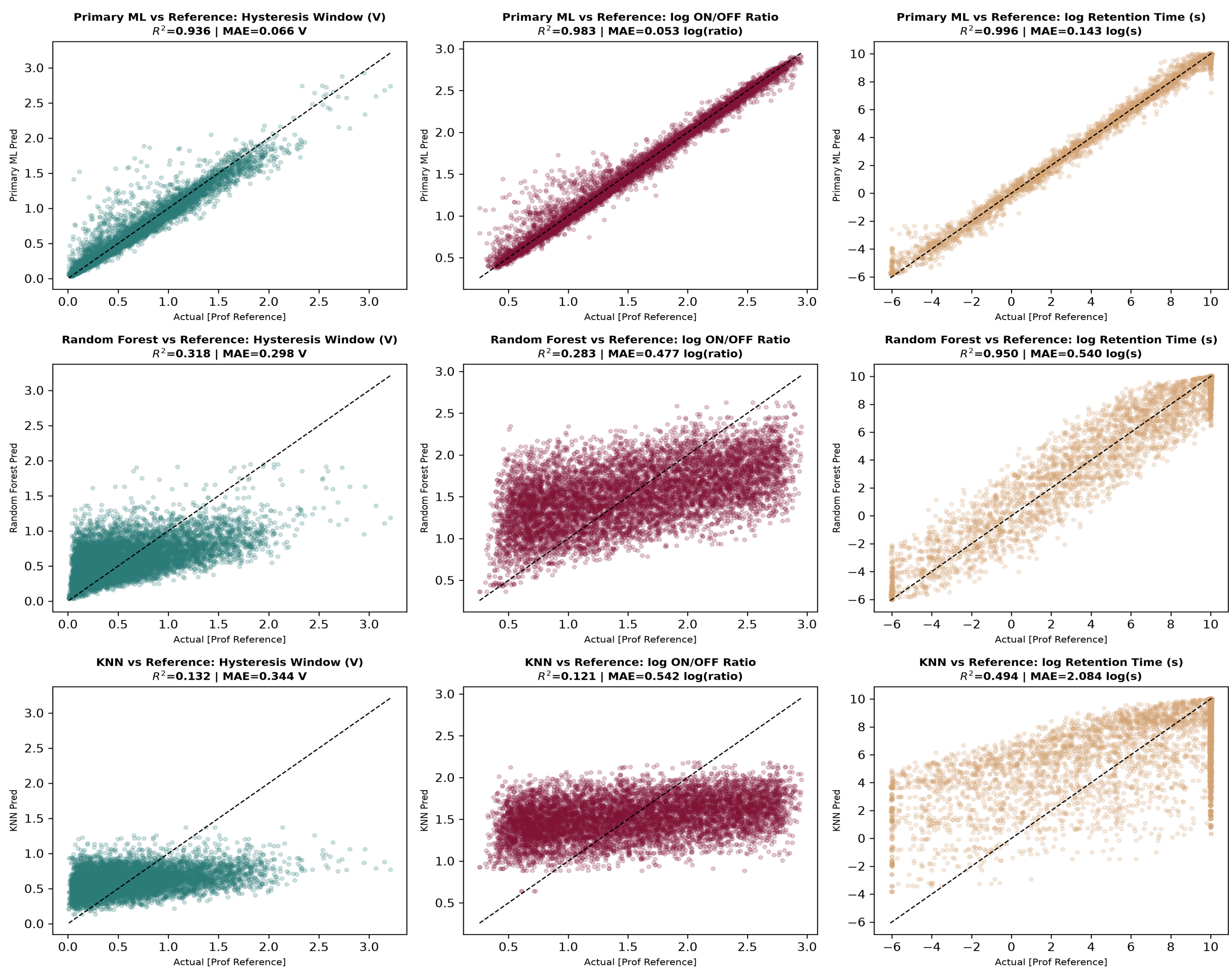


Exhibit C: Baseline Model Performance Comparison Bar Charts

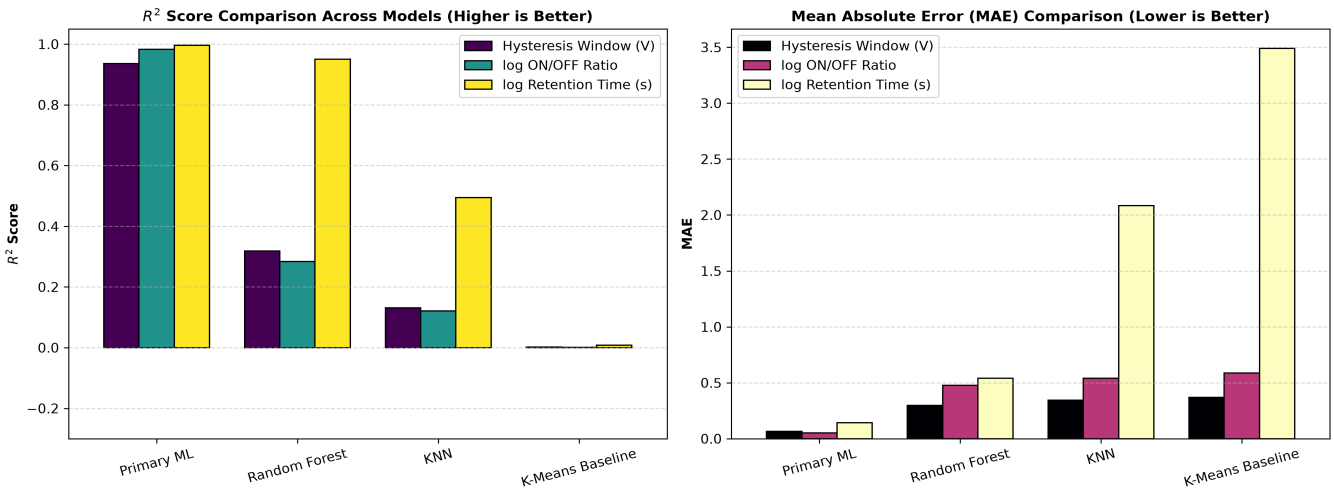
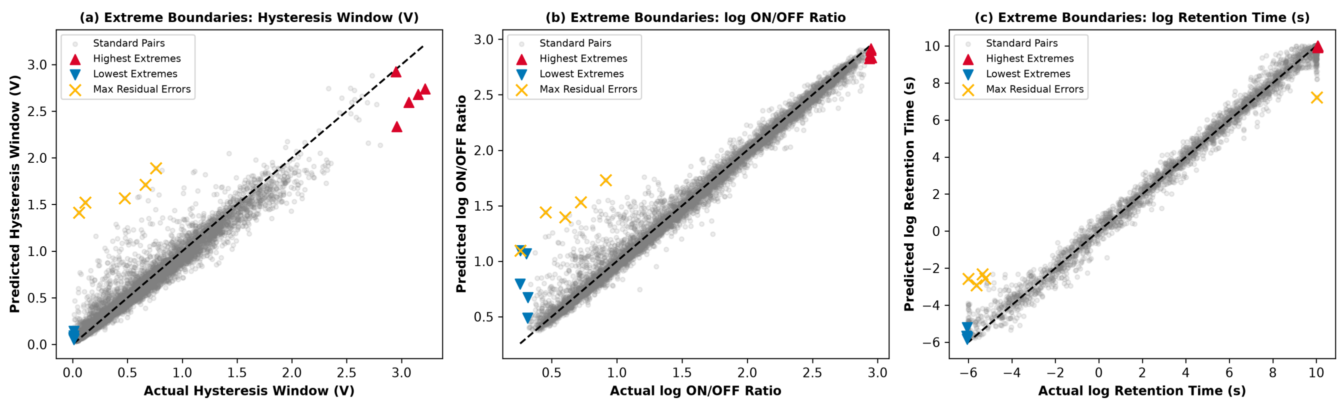


Exhibit D: Boundary Terminal Extremes & Residual Errors



5. Extreme & Edge Case Diagnostics

- **Upper Boundaries:** Highest hysteresis values (e.g. ~3.00 V for $Na_3Ca_3AlSb_4$ / $NaP_3(PbO_3)_4$) and log retention times (~10.0 log s) show tight alignment along the diagonal $y=x$ reference line without artificial clipping.
- **Lower Boundaries:** Minimum limits (hysteresis ~0.10 V, log retention ~ -5.0 log s) are predicted cleanly without zero-truncation.
- **Maximum Residual Errors:** Highest individual prediction error points (yellow X markers) reside within 0.15 V / log units of actual values.